Response to reviewers for

Manuscript Title:
Analysis of human visual experience data

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We thank the editor and the reviewers for their insights and feedback. We have made several changes to the manuscript based on this feedback. We reference these changes point by point, together with the reviewer's remarks. Reviewers comments are written verbatim, with our point by point replies in **bold**.

Reviewer 1:

I am mostly satisfied with how the authors' addressed my comments to the initial manuscript, and I approve of the changes made to the manuscript.

One issue that remains is the absence of background information on the behaviours of the participants whose data are presented in the examples. My original review commented that:

"My other complaint with the manuscript was how the tutorial handled the behavioural data of the participants. I am shown code which handles periods of "visual breaks", "illuminance distribution", "average light exposure", etc, but no comment on whether the results of these analyses correspond to the sensor wearer's actual behaviour. Why, for example, does the participant in Table 9 have less mean daylight exposure at the weekend than during week days?"

The authors address this by replying:

"We agree that the individual results deviate quite a bit from typical values seen in groups."

and do, indeed, include a disclaimer in the manuscript. This is a satisfactory response and I don't see the need to demand a resubmission, but I feel it does not fully address my original objection.

Manuscripts like this one (which are demonstrative in nature, and are not bound to any particular data set) should include data sets where the reader has a chance to reason whether the presented analysis is correct or not. Had the example data included a typical school child on a typical school day, then the reader would have an expectation of typical daylight/indoor-light exposure patterns (since we all have experience of school days, indoor lessons, outdoor recesses, etc) and would be able to assess whether LightLogR had correctly captured these patterns. Instead, the authors show some arbitrary (and, apparently atypical) data with no context of what was happening during the measurement period. The authors have (inadvertently, I hope) introduced an additional layer of abstraction here -- the reader now has no expectation of outcomes and is unable to 'eye ball' the output of LightLogR and assess its correctness. This extra abstraction feels unnecessary.

May I politely request the authors consider adding some brief explanation of the participant's behaviours to the final manuscript. Such explanations would help address "Why, for example, does the participant in Table 9 have less mean daylight exposure at the weekend than during week days?", as per my original review. Was the lower mean lux at the weekend merely due to cloudy weather? Less time spent outdoors than weekdays? A bug in LightLogR? An error transposing results to the manuscript? Please give me (the reader) a fighting chance to understand the analysis behind this table!

Our reply:

We thank Reviewer 1 for the encouragement and constructive feedback. We agree that this issue is a hurdle to the general uptake of the content. The data were from a sample collected during COVID, which meant students spent an unusual amount of time in indoor environments. We have searched for another dataset from the same pool that has more common values for illuminance and have added a citation and some information about the data both in the supplement and the main manuscript (under #Devices). These changes affect all analyses that work with the ClouClip data. In two cases, we had to adjust the metric definition in order to showcase metrics for that individual. For duration of continuous near work, the minimum duration had to be reduced, as this person would have otherwise had no continuous near work episodes. For visual breaks, a minimum near-work duration before the visual break (1 minute) was necessary, as this individual had many measurements around the threshold of 100 cm, which would have artificially increased the number of breaks. We have expanded on the flexibility of the implementation in the caption of Table 2.

Reviewer 1:
I have the following minor comments about the updated manuscript:

Line 212: "The package further supports versions due to..." - I believe the correct software enginering term here is "versioning". Is that what you meant?

Our reply:

That is correct. We have adjusted the sentence to read “The package further supports versioning due to evolving data formats […]”

Reviewer 1:
Table 14 would benefit from adding "lux" (or "lx") to the column headings

Table 17 would benefit from adding "mW/m^2" to the "short" and "long" column headings

Table 18 would benefit from adding "lux" (or "lx") to the "illuminance" column heading

Table 20 would benefit from adding "mW/m^2" to the "day" and "night" column headings

Our reply:

We have added these suggestions to the column headings

Reviewer 2:
I thank authors for responding to reviewer comments and feedback, the manuscript has significantly improved and easier to read. I only have minor comments which refer to the tracked version of the manuscript.

Our reply:

We thank Reviewer 2 for the encouragement and feedback.

Reviewer 2:
Table 2: apologies if I have missed this but is there opportunity to modify some of the proposed definitions to suit the researcher which I believe is the case looking at the later coding. It may be worthwhile to include this somewhere early in the manuscript such as in the Metrics selection and definitions section
Our reply:

Yes, it is possible to adjust both thresholds and also the number of bins. We have added a sentence towards this in the caption of Table 2

Reviewer 2:
Line 224: when rounding timestamps to the nearest regular interval, what happens if there is already a data point there? does it get averaged?
Our reply:

That method is only feasible if no duplicate time points result, otherwise aggregation should be used. We have clarified that in the manuscript.

Reviewer 2:
Line 249: was the same procedure applied to Clouclip data, where days with more

than 1 hour of missing illuminance removed?
Our reply:

A similar procedure was used, but instead of dropping days with more than 1 hour missing, we dropped days with less than 1 hour remaining. This is written 2 paragraphs above where it reads:

"In our Clouclip example, days with less than one hour of recordings were dropped. This threshold should be adjusted based on how much complete days matter for a given analysis at hand. E.g., in circadian science, the metrics of interdaily stability and intradaily variation require measurements for each hour of the day."

Reviewer 2:
Line 969: Authors have also acknowledged that fleeting spikes about 1000 lux may not represent meaningful outdoor exposures and proposed that sustained measures above 1000 lux may be able to better capture true transition. I'm wondering if another approach could be looking at greater lux levels changes from above 1000 to under 950 lux, as a example?
Our reply:

In principle, yes. In our experience, the difference between indoor and outdoor conditions is much stronger (see bimodal distribution in Figure 8, Histogram), so we would suggest changes from above 1000 to under say 600 lx. There are many ways how to operationalize the idea of the transition. We have expanded on this in the text and Figure 10.